

MANJUNATH M V

ECE STUDENT | ASPIRING ANALOG IC & LAYOUT DESIGN ENGINEER

Analog IC Design | VLSI Design | Layout Design | Semiconductor Engineering

Motivated Electronics & Communication Engineering student passionate about Analog IC Design, Layout Design and VLSI Systems. Hands-on experience with Cadence Virtuoso, SRAM design, analog circuit design, and semiconductor technologies through academic and personal projects. Seeking internship and entry-level opportunities to apply my skills and contribute to innovative chip solutions.



manjunathvannur21@gmail.com

Bengaluru, India

Graduate in 2027

MY STATS



8.14

CGPA



6+

PROJECTS



10+

CERTIFICATIONS



2027

GRADUATE

TECHNICAL SKILLS

ANALOG DESIGN

- Two-Stage CMOS Op-Amp
- Current Mirrors
- Differential Amplifiers
- CMOS Design
- Analog Circuit Analysis

LAYOUT DESIGN

- Cadence Virtuoso
- 180nm & 90nm Technology
- Layout Planning
- DRC, LVS

VLSI DESIGN

- SRAM Design
- SRAM I/O Circuitry
- RTL Design (Verilog)
- Low Power Design

SIMULATION TOOLS

- Cadence Virtuoso
- LTspice
- ModelSim
- COMSOL Multiphysics
- MATLAB

PROGRAMMING

- Verilog HDL
- C / C++
- Python

HARDWARE PLATFORMS

- ESP32
- Arduino Nano
- Arduino IDE

CURRENTLY LEARNING

- Analog Layout Design
- CMOS Analog Circuits
- Static Timing Analysis (STA)
- Advanced SRAM Design
- Physical Design Fundamentals
- High-Speed Circuit Design



ABOUT ME

Electronics & Communication Engineering student at Dayananda Sagar College of Engineering (DSCE), Bengaluru, graduating in 2027. Passionate about Analog IC Design, Layout Design, SRAM Design, and VLSI Systems. Experienced with Cadence Virtuoso (180nm & 90nm), LTspice, Verilog HDL, ModelSim, MATLAB, and COMSOL through academic and personal projects. Seeking internship and entry-level opportunities to apply semiconductor design skills and contribute to innovative silicon solutions.



FEATURED PROJECTS

1

SRAM I/O Circuit Design & Verification

Designed and verified SRAM I/O circuits including Write Driver, Sense Amplifier and Precharge Circuit using Cadence Virtuoso (180nm). Performed extensive simulations and ensured functional & timing correctness.

Cadence Virtuoso

2

DFF Optimized Low Power SRAM for Satellite Systems

Designed a low power SRAM using DFF architecture for satellite applications in 180nm. Achieved improvements in power, stability and read/write reliability.

Cadence Virtuoso

3

Two Stage CMOS Operational Amplifier

Designed and simulated a two-stage CMOS op-amp with high gain and improved stability in 180nm. Performed DC, AC, Transient and Corner simulations.

Cadence Virtuoso

4

MedShield AI

Developed an AI-powered security system using YOLOv8, DeepFace and ESP32-CAM for real-time threat detection and alerting.

Python, ESP32

5

All Optical Plasmonic Switch using GST

Modeled and analyzed a $\text{Ge}_2\text{Sb}_2\text{Te}_3$ based all-optical plasmonic switch using COMSOL Multiphysics to evaluate performance in photonic applications.

COMSOL

6

Smart Shopping Cart System

IoT based smart cart with automatic billing, object detection and real-time display using Arduino Nano and sensors.

Arduino, C++



CERTIFICATIONS

VLSI & SEMICONDUCTOR

- RTL to GDS II Masterclass
- NPTEL Analog Circuits (61%)
- VLSI Made Easy Webinar
- SCL PDK Training

TOOLS & SIMULATION

- MATLAB Onramp
- MATLAB Signal Processing

EMBEDDED & PROGRAMMING

- Embedded System Design
- Using C

EMERGING TECHNOLOGIES

- Generative & Agentic AI
- Internet Society CCN



ACHIEVEMENTS

Winner – Circuit Minds Theme
TechnoCognition Hackathon 2025
for "DFF Optimized Low Power SRAM for Satellite Systems"



Active participant in technical workshops, webinars and hackathons.



TIMELINE

2023

2025

2025

Nov 2025

Jan – Mar 2026

Feb 2026

Feb 2026

2026

May 2026

B.E. ECE at Dayananda Sagar College of Engineering (DSCE), Bengaluru
Smart Shopping Cart System
All Optical Plasmonic Switch using GST (COMSOL)
MedShield AI
DFF Optimized Low Power SRAM for Satellite Systems
NPTEL Analog Circuits
RTL to GDS II Training
Two Stage CMOS Op-Amp Design
SRAM I/O Circuit Design & Verification